A mini course in meta-analysis

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Session contents

- Introduction to meta-analysis
- R tutorial and statistics of meta-analysis
- Discussion of some “advanced” topics in meta-analysis

Meta-analysis

Meta-analysis: A statistical technique for combining results from multiple studies on a similar topic to obtain a quantitative summary estimate of the effect size.

(why do it? what does "similar" mean? how to combine?)

Why meta-analysis

1. Provides a more comprehensive and precise estimate of the effect size by pooling data from multiple studies.
2. Increases generalizability of results.
3. Helps to resolve inconsistencies or contradictions in individual studies.
4. Allows for exploration of heterogeneity and subgroup analyses.

Tension between 1 and 2: can you always be more precise and more generalisable?

Conducting a meta-analysis

1. Define research question and objectives.
2. Conduct a systematic literature review to identify relevant studies.
3. Extract data from each study, including sample size, effect size estimates, and study characteristics.
4. Perform statistical analyses, including calculation of effect size estimates, pooling of effect sizes using appropriate statistical methods, and assessment of heterogeneity.
5. Interpret the results and draw conclusions.

Alongside: conduct sensitivity analyses and assess publication bias.

Effect size

- Interpretable
- Reliable
- Comparable

No consistent definition of effect size. “Comparable” is subjective. Consider ivermectin example.

Typical measures: - Means, proportions, correlations - Mean differences, risk and odds ratios, IRRs, HRs, . . .

Heterogeneity: variation in effect sizes

1. Heterogeneity: The variability of effect sizes across different studies included in a meta-analysis
2. Sources of variation: “clinical”, methodological, and statistical
3. Heterogeneity can impact the validity and generalizability of meta-analysis results
4. Statistical methods, such as the Q statistic and I^2 statistic, can be used to assess the extent of heterogeneity.

Internal and external validity

Quick reminder:

- Internal validity: Validity of findings within individual studies
- External validity: Generalizability of findings beyond the original studies

Meta-analysis relies on the assumption of internal validity, as the results within each study included are assumed to be reliable and unbiased. Assessing external validity is important to determine the scope in which results are applicable. *Choice of model depends on objective.*

Fixed-Effect Meta-Analysis

1. Definition: A meta-analysis model that assumes a single true effect size across all studies.
2. Assumes that any observed differences between studies are due to random error.
3. Typically used when there is little or no heterogeneity among studies.
4. Pooled effect size estimate is calculated using weighted averages of individual study effect sizes, with weights based on sample size or other factors.

Random-Effects Meta-Analysis

1. Definition: A meta-analysis model that accounts for both within-study and between-study variability in effect sizes.
2. Assumes that there may be multiple true effect sizes across studies, due to both random error and systematic differences.
3. Typically used when there is significant heterogeneity among studies.
4. Pooled effect size estimate is calculated using weighted averages of individual study effect sizes, with weights based on both sample size and estimated between-study variability.

This is simplified on purpose!

Meta-regression

1. Meta-regression is a statistical technique used in meta-analysis to explore the potential sources of heterogeneity among studies.
2. Allows for the examination of how different study-level characteristics (e.g., age, gender, publication year) may influence the overall effect size estimate.
3. Conducted using regression: effect sizes from each study are regressed against the study-level covariates
4. Meta-regression can provide valuable insights into the potential sources of heterogeneity and help explain variability in effect sizes among studies.

It's a simple technique. Applicable to both FE and RE models. Many of you know it as a mixed model.

Reporting meta-analysis results

(put in an image here)

We will examine more examples soon.

Implementing meta-analyses

This session is generic. However:

1. R is a good tool for meta-analyses. Reasons:
   - Meta-analyses are not computationally intensive
   - A lot of data processing, visualisation, need for specialised ecosystem (do not code your own solutions!)
   - Big focus on reporting to non-technical audiences
2. General R packages: `meta`, `metafor`, `baggr`
3. Each of them has good tutorials (look online or in `vignettes()` in R)

```r
library(meta)
# Import data from CSV file
data <- read.csv("meta_data.csv")
# Compute effect sizes and standard errors
effect_size <- log(data$odds_ratio)
# Typical way of approximating SE(log(OR)), we will discuss this later
se <- sqrt(1/data$a + 1/data$b + 1/data$c + 1/data$d)
# Create meta-analysis object
meta_result <- meta::metagen(effect_size, se, studlab = data$study_name)
# Forest plot
meta::forest(meta_result, xlim = c(-5, 3), xlab = "Log Odds Ratio",
             atransf = exp, cex = 0.8,
             print.tau2 = FALSE,
             comb.random = TRUE,
             comb.fixed = TRUE)
# Meta-regression (if applicable)
meta::metareg(meta_result, ~covariate1 + covariate2)
# Funnel plot (if applicable)
meta::funnel(meta_result)
```

Conclusion to this intro

Terms we introduced:

- meta-analysis,
- effect size,
- generalisability,
- heterogeneity,
- fixed effects,
- random effects,
- meta-regression,
- forest plot

Questions?